

Web Development (WD)

✓ What is Data ?

- ✓ Data means Collection of characters ,text or information.
- ✓ Data is a collection of facts, information, and statistics and this can be in various forms such as numbers, text, sound, images, or any other format.

✓ Webpage :-

- Webpage is a collection of data.
- A webpage is a single document on the internet.
- It is written in HTML (Hyper Text Markup Language).

✓ Website :-

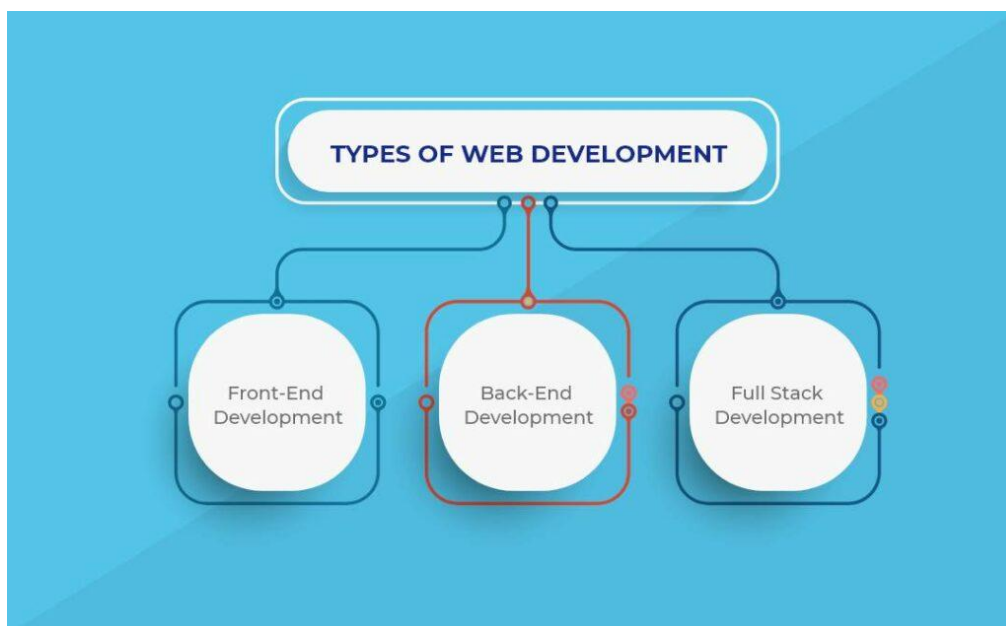
- Website is a collection of related webpages.
- A website is a collection of related webpages.
- These webpages are linked together through hyperlinks.

✓ Web Development :-

- Web development means creating and maintaining websites.
- It involves building webpages and web applications.

✓ Types of Web Development :-

Web development is divided into two main parts: Frontend and Backend.



1) Front-End Development :-

- Frontend development is the part of web development that deals with what the user sees.
- The main technologies used are HTML, CSS, and JavaScript.
- Frontend developers often use frameworks and libraries like React, Angular, Vue.js, Bootstrap.

2) Back-End Development :-

- Backend development is the part of web development that works behind the scenes.
- It uses programming languages like PHP, Python, Java, Node.js, Ruby, C#.
- The backend also uses databases like MySQL, MongoDB, PostgreSQL, and Oracle.

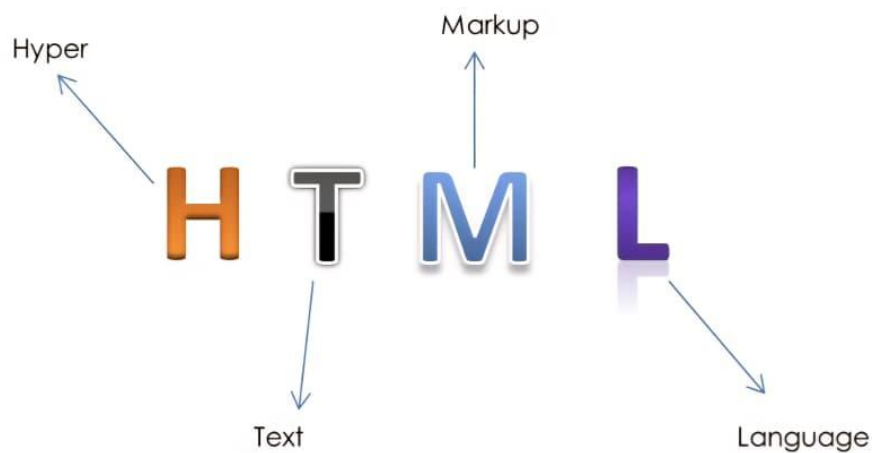
3) Full Stack Development :-

- Full Stack Development means working on both the frontend and backend of a web application.
- The Frontend is what users see in the browser (UI/UX part).
- It uses technologies like HTML, CSS, JavaScript, and frameworks like React, Angular, Vue.js.
- The Backend is what happens behind the scenes (server, logic, database).
- It uses programming languages like Node.js, PHP, Python, Java, Ruby, C#.
- Backend also manages databases such as MySQL, MongoDB, PostgreSQL, Oracle.

HTML (Hyper Text Markup Language)

✓ What is HTML ?

- HTML stands for HyperText Markup Language.
- HTML is used to create web pages and web applications.
- We can create a static website by HTML only.
- HTML is a Markup language rather than a programming language.



✓ Boilerplate code of HTML :-

```
<html>  
  <head>  
    <title>Website Title</title>  
  </head>  
  <body>  
    <h1>Hello Geeks</h1>  
    <h2>HTML Basic Structure</h2>  
    <p>Paragraph Tag</p>  
  </body>  
</html>
```

Where,

- `<!DOCTYPE html>` → Document type (HTML5).
- `<html>` → Root tag.
- `<head>` → Metadata (title, styles, scripts).
- `<body>` → Visible content of the webpage.

✓ **Tags in HTML :-**

- A tag in HTML is a special keyword written inside angle brackets (`<` `>`).
- Tags are the building blocks of an HTML document.
- They tell the browser how to display content.
- Most tags come in pairs: an opening tag and a closing tag.

✓ **Types of Tags :-**

HTML Tags can be divided into various types based on different aspects :

1) Paired Tag –

- It is a combination of opening tag & closing tag.
- The opening tag looks like `<tagname>` and the closing tag looks like `</tagname>`.
- Content is written between the opening and closing tags.

Example –

- `<html> ... </html>`
- `<head> ... </head>`
- `<body> ... </body>`
- `<h1> ... </h1>`
- `<p> ... </p>`
- `<div> ... </div>`

2) Unpaired/ Unclosed Tag -

- It is a self closing tag no need to close manually.
- These tags do not contain any content inside them.
- They are mostly used for line breaks, images, or rules.

Example –

- `<br>` → inserts a line break
- `<hr>` → inserts a horizontal line
- `` → displays an image